SOLID STATE

1. Refractive index of a solid is observed to have the same value along all directions. It means nature of solid is–

(1) Anisotropic (2) Isotropic
(3) Crystalline (4) None of these

2. Quartz glass is a

(1) Amorphous Solid (2) Crystalline Solid
(3) Anisotropic (4) None of these

3. Diamond is a

(1) Molecular solid (2) Ionic solid
(3) Metallic solid (4) Covalent solid

4. Which type of solids are electrical conductors, malleable and ductile?

(1) Molecular solid (2) Metallic solid
(3) Ionic solid (4) covalent solid

5. H_3BO_3 is an example of which crystal system?

(1) Cubic (2) Tetragonal
(3) Triclinic (4) Monoclinic

6. A compound forms hexagonal close packed structure. How many octahedral voids in 0.3 mol of it?

(1) 1.08×10^{23} (2) 1.80×10^{23}
(3) 5.4×10^{23} (4) None of these

7. A compound is formed by three elements A, B and C. The element A forms CCP, atoms of B occupy $2/3^{rd}$ of tetrahedral voids & atoms of C occupy $1/3^{rd}$ of octahedral voids then formula of the compound is–

(1) ABC (2) $A_3B_4C_3$
(3) A_3B_4C (4) A_4B_3C

8. The two dimensional Co-ordination number of a molecule in square closed-packed layer is–

(1) 6 (2) 8
(3) 12 (4) 4

9. For graphite which is possible?

(1) $a = b = c,\ \alpha = \beta = \gamma = 90°$
(2) $a = b \neq c,\ \alpha = \beta = \gamma = 90°$
(3) $a = b \neq c,\ \alpha = \beta = 90°,\ \gamma = 120°$
(4) $a \neq b \neq c,\ \alpha = \beta = \gamma = 90°$

10. In dislocation defect, density of solid is–

(1) increased (2) decreased
(3) unchanged (4) None

11. Co-ordination number in Cu metallic crystal is–

(1) 6 (2) 8
(3) 12 (4) 4

12. Volume occupied by atoms in FCC unit cell is–

(1) $\frac{4}{3}\pi r^3$ (2) $\frac{8}{3}\pi r^3$
(3) $\frac{16}{3}\pi r^3$ (4) πr^3

13. Due to excess of lithium. It makes LiCl crystals–

(1) yellow (2) violet
(3) pink (4) white

14. Solid solution of $CdCl_2$ and AgCl show which type of defect?

(1) Schottky (2) frenkel
(3) Impurity (4) metal excess

15. A metal crystallises in FCC lattice. If edge length of the cell is 4×10^{-8} cm and density is 10.5 g cm^{-3} then the atomic mass of metal is–

(1) 101.2 (2) 50.6
(3) 202.3 (4) None